Information Required from Videos

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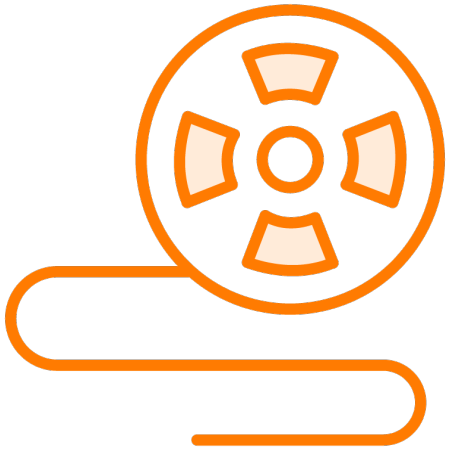
Video Analysis Overview



When looking at images there were many types of feature available.

Videos are sequences of images with audio.

There are vast amounts of information that may be of interest.

Videos introduce new types of analysis.

The Azure AI Video Indexer contains numerous AI models that provide video insight.

Can also be used to provide deep search capabilities over a video library.

Key Types of Video Analysis

- Transcription and enhancements
- Face and people identification
- OCR
- Scene segmentation and shot detection
- Content moderation
- Topic and sentiment

Using Azure Video Indexer

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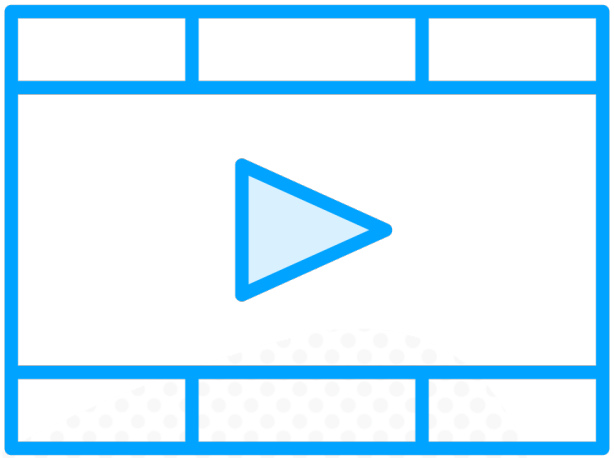
Azure AI Video Indexer Resource

This is a separate type of resource.

A free instance is created by default with limited capabilities.

Create an instance for full capabilities and customization.

Features are not part of the multi-service account.



Available Indexing Bundles

Basic	Standard	Advanced
Audio: Transcription, translation, formatting of output captions and subtitles (closed captions) Video: Labels, object detection, OCR, scenes (keyframes and shots), and black frame detection.	Audio: + Automatic language detection, emotions, keywords, named entities (brands, locations, people), sentiments, speakers, topic extraction, and textual content moderation Video: + Named entities (OCR - brands, locations, people), people, visual content moderation, and topic extraction (OCR).	Audio: + Audio event detection Video: + Matched person, observed people, clapperboard detection, digital pattern detection, featured clothing insight, textless slate detection, textual logo detection

While the specific analysis performed can be customized you pay based on the “bundle” selected.

Customizing Analysis

Customization is available to enhance analysis on tailored requirements.

- Speech – Enables recognition of domain-specific vocabulary, terms and/or pronunciation
- Language ID – Improve language auto-detection
- People – Identify unique people
- Brand – Find additional brands

Demo: Using the Portal

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Key Steps

Uploading a video

Selecting analysis options

Viewing the analysis

Using Widgets

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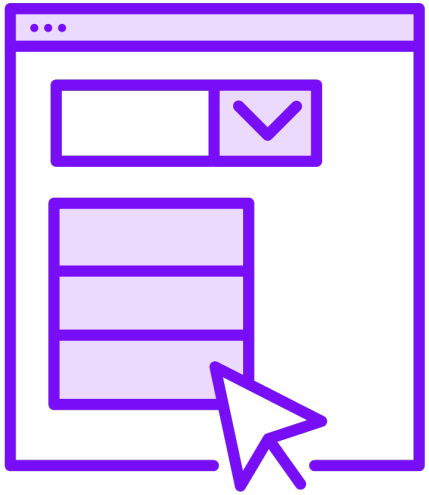
Widget Capabilities and Usage



Widgets expose elements of Azure Video Indexer to your own applications and sites.

The embedded code can be generated by the portal or manually created.

Commonly you will leverage:

- Player – Viewing the video
- Insights – View insights about the video

If using both widgets for a video a mediator script is required to enable interaction between them.

Automating Indexing Tasks

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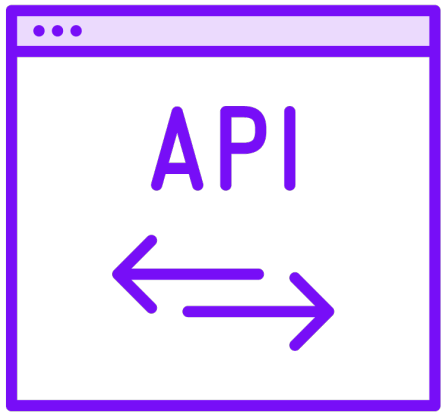
Using the Azure Video Indexer API



Often tasks need to be automated which is possible via the provided API.

There are currently no SDKs available.

There is a separate developer portal available that provides guidance on using the API.

https://api-portal.videoindexer.ai/

Access tokens are required to perform operations and last 1-hour.

Access tokens for the Operations API are acquired via ARM (or the Authorization API).

Analyzing Videos Review

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Summary

Videos have many different types of analysis available.

When performing analysis you select on a per-video basis:

- Audio, video or audio & video
- The bundle of analysis performed

Pricing is based on minutes analyzed and the level of analysis.

Analysis insights can be viewed using the portal, widgets and via the API.

The API can also be used to automate all the types of video analyzer operation.